# SAC-Net: Symmetry-Aware Cascaded Attention for Panoramic Tooth Segmentation

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ABSTRACT

Accurate tooth segmentation from panoramic dental radiographs is a fundamental task in computer-aided dental diagnosis. Standard segmentation architectures often fail to use the inherent anatomical symmetry of the dental arch, where teeth exhibit mirror relationships across the midline, leading to ambiguous predictions for visually similar categories. To address this, we propose SAC-Net, a novel symmetry-aware architecture built upon a U-Net backbone. Our approach introduces two core components: the Symmetric Cross-Midline Attention (SCMA) module, which establishes bidirectional feature modulation between corresponding regions across the arch, and the Cascaded Symmetric Refinement Block (CSRB), which iteratively refines features by enforcing global symmetry constraints while preserving local details. These modules are integrated at multiple encoder stages and the network bottleneck, creating a cascade of symmetry-enforcing operations throughout the network. Validated on the publicly available STS-Tooth dataset comprising 850 annotated panoramic X-ray images, SAC-Net shows superior segmentation performance by explicitly modeling cross-arch contextual relationships, effectively utilizing the structural prior of the dental arch for robust feature learning. The code and checkpoints are available at https://github.com/yifangao112/SAC-Net

## 1. Introduction

Accurate tooth segmentation from panoramic dental radiographs is a fundamental prerequisite for computer-aided diagnosis and treatment planning in modern dentistry Shaker, Maaz, Rasheed, Khan, Yang, and Khan (2022). It enables automated detection of dental pathologies, assessment of skeletal anomalies, and planning for orthodontic and implant procedures Chen, Ma, Xu, and Xu (2024). While deep learning, particularly convolutional neural networks (CNNs) based on the U-Net architecture Oktay, Schlemper, Folgoc, Lee, Heinrich, Misawa, Mori, McDonagh, Hammerla, and Kainz (2018), has achieved significant progress in medical image segmentation, the unique anatomical structure of the dental arch presents a persistent challenge that standard architectures fail to address effectively.

The human dentition exhibits a pronounced bilateral symmetry, where teeth on the left and right sides of the midline are approximate mirror images of each other. This structural prior is a powerful cue for disambiguating visually similar tooth categories and resolving segmentation ambiguities, especially in regions with low contrast, overlapping structures, or pathological alterations. However, conventional semantic segmentation models treat spatial locations independently, processing the entire image without explicitly encoding this inherent symmetry. Consequently, they learn features in isolation, missing the opportunity to use the consistent cross-arch relationships that could regularize and refine the segmentation process. Existing approaches that attempt to incorporate symmetry often do so through heuristic, post-processing steps applied after network inference, which are not integrated into the feature learning pipeline and cannot provide end-to-end optimization Seo, Kim, Kwak, and Cho (2022).

Modern attention mechanisms, while powerful for capturing long-range dependencies, typically operate asymmetrically across the entire feature map. They lack a built-in inductive bias to specifically model the bidirectional correspondence between symmetric anatomical regions across the dental midline. This results in suboptimal feature representations where the left and right arch features are refined in isolation, rather than through mutual contextual exchange. The absence of a dedicated mechanism to enforce this anatomical

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prior during training limits model performance, particularly in achieving precise boundary delineation, which is clinically critical.

To overcome these limitations, we propose SAC-Net, a novel Symmetry-Aware Cascaded attention network for panoramic tooth segmentation. Our core contribution is the explicit modeling of cross-arch contextual relationships through two cascaded, symmetry-enforcing modules integrated into a U-Net backbone. First, we introduce the Symmetric Cross-Midline Attention (SCMA) module, a lightweight bidirectional feature modulation mechanism that establishes explicit communication between corresponding left and right arch regions at multiple encoder stages. This allows features from one side to directly refine and disambiguate features on the opposite side during the hierarchical feature extraction process. Second, we design a Cascaded Symmetric Refinement Block (CSRB) placed at the network bottleneck. The CSRB iteratively applies symmetric alignment, local detail enhancement, and channel recalibration in a cascaded manner, enforcing global symmetry constraints on the most abstract feature representations while preserving fine-grained anatomical details.

Extensive experiments on the publicly available STS-Tooth dataset, comprising 850 annotated panoramic X-ray images, show the effectiveness of our approach. SAC-Net achieves state-of-the-art performance with a Dice score of 0.9416 and a Hausdorff Distance (95th percentile) of 0.7943, outperforming strong contemporary baselines including U-Mamba, UKAN, and nnU-Net. Crucially, our model maintains competitive computational efficiency with 18.18M parameters and an inference time of 4.3 ms. Ablation studies confirm the contribution of each proposed component, and qualitative results show that SAC-Net produces segmentations with more accurate boundaries and fewer false positives, especially for challenging cases near the arch midline. By embedding the anatomical prior of dental symmetry directly into the network architecture, our work provides a principled framework for using domain-specific structural knowledge to advance the accuracy and robustness of medical image segmentation.

## 2. Related Work

### 2.1. Medical Image Segmentation Architectures

The evolution of medical image segmentation architectures has been largely defined by the encoder-decoder paradigm, which excels at capturing hierarchical features and recovering spatial resolution. The foundational Fully Convolutional Network (FCN) Shelhamer, Long, and Darrell (2014) showed that convolutional networks could produce dense pixel-wise predictions, establishing the template for semantic segmentation. This design was significantly advanced by the U-Net architecture Oktay et al. (2018), which introduced symmetric skip connections between the contracting encoder and expanding decoder to preserve fine-grained spatial details lost during downsampling. The U-Net's efficiency and effectiveness with limited annotated data made it a de facto standard for biomedical imaging. Subsequent variants sought to enhance its representational capacity and connectivity. The DeepLab series Chen, Zhu, Papandreou, Schroff, and Adam (2018) incorporated atrous spatial pyramid pooling to capture multi-scale context, while DenseNet-based designs Jégou, Drozdzal, Vázquez, Romero, and Bengio (2016) promoted feature reuse through dense connections within the encoder-decoder blocks. The nnU-Net framework further systematized this paradigm by showing that a robustly configured and trained vanilla U-Net could achieve state-of-the-art results across diverse challenges, emphasizing the critical role of tailored preprocessing and postprocessing over architectural novelty alone.

To address the limited receptive field of standard convolutions and model long-range dependencies, attention mechanisms were integrated into segmentation networks. The Attention U-Net Oktay et al. (2018) introduced a spatial attention gate within skip connections, allowing the decoder to focus on salient regions and suppress irrelevant feature responses from the encoder. This represented a shift from purely local processing to content-aware feature aggregation. Concurrently, the squeeze-and-excitation network Hu, Shen, Albanie, Sun, and Wu (2017) proposed channel-wise attention to recalibrate feature maps by modeling interdependencies between channels. These self-attention mechanisms, however, were typically applied locally or within a limited spatial context. A more radical departure arrived with the Vision Transformer (ViT), which replaced convolutional inductive biases entirely with a sequence-based self-attention mechanism, demonstrating remarkable performance on large-scale image classification by capturing global context. For dense prediction tasks, models like SegFormer Xie, Wang, Yu, Anandkumar, Álvarez, and Luo (2021) adapted

the Transformer to segmentation by designing a hierarchical encoder and a lightweight all-MLP decoder, efficiently generating high-resolution outputs.

The pursuit of architectures that combine the local feature precision of CNNs with the global modeling capacity of Transformers led to the development of hybrid models. TransUNet Chen, Lu, Yu, Luo, Adeli, Wang, Lu, Yuille, and Zhou (2021) was a pioneering hybrid that used a CNN to extract low-level spatial features and a Transformer at the bottleneck to model global context, bridging the two paradigms effectively. This inspired numerous successors, including SwinUNETR He, Nath, Yang, Tang, Myronenko, and Xu (2023), which integrated a hierarchical Swin Transformer backbone with a U-Net decoder for 3D medical image analysis. More recently, architectures based on state-space models, such as U-Mamba Lei and Yin (2024), have emerged as efficient alternatives for long-sequence modeling, offering linear complexity with performance competitive to Transformers. Furthermore, diffusion models have been adapted for segmentation, framing it as a conditional denoising process to generate high-quality segmentation masks Baranchuk, Rubachev, Voynov, Khrulkov, and Babenko (2021). Despite these advances, a common limitation persists across these diverse architectures: they treat spatial locations largely independently or aggregate context in a symmetric, non-discriminative manner. This design fails to encode powerful, domain-specific structural priors, such as the bilateral symmetry inherent in anatomical structures like the dental arch Oktay et al. (2018) Chen et al. (2021). Consequently, existing methods either neglect such priors or apply them through heuristic, post-hoc corrections that are not integrated into the end-to-end feature learning process, leaving a gap for architectures that can explicitly and learnably enforce anatomical constraints to resolve ambiguities in challenging segmentation tasks.

## 2.2. Symmetry Modeling in Vision Tasks

Early approaches to symmetry modeling in computer vision relied on explicit geometric constraints and post-processing heuristics. These methods typically detect symmetry axes or correspondences in the input image or the model's output, then enforce consistency through alignment or regularization. For instance, works on facial landmark detection and object pose estimation often incorporate symmetry as a geometric loss that penalizes deviations between predicted left and right keypoints Trigeorgis, Snape, Nicolaou, Antonakos, and Zafeiriou (2016) Zhou and Tuzel (2017). In medical imaging, similar post-hoc regularization has been applied to enforce symmetry in brain hemisphere segmentation or cardiac structure delineation Raina, Yahorau, and Schmah (2019). While straightforward to implement, these methods operate as a separate, often non-differentiable stage, decoupled from the feature learning process. Consequently, they cannot guide the network to learn intrinsically symmetric representations and may fail when the underlying anatomy exhibits significant non-rigid variations or when the initial predictions are too erroneous for the post-processing step to correct Chen and He (2020).

A more integrated strategy involves architectural priors that bake symmetry into the network's forward pass. This category includes weight-sharing mechanisms like Siamese networks for learning symmetric feature embeddings Chang and Chen (2018) and, more fundamentally, group-equivariant convolutional networks (G-CNNs). G-CNNs use specialized convolution kernels designed to be equivariant to predefined symmetry groups, such as rotations or reflections, ensuring that feature transformations commute with group actions Cohen, Geiger, and Weiler (2018) Worrall, Garbin, Turmukhambetov, and Brostow (2016). Extensions and applications of this principle have shown promise in molecular property prediction and galaxy morphology classification where symmetries are exact Cohen, Geiger, and Weiler (2018) Dieleman, Willett, and Dambre (2015). However, a core limitation of strict architectural equivariance is its assumption of perfect, global symmetry. In biological systems like the human body, symmetry is approximate and locally variable; enforcing hard equivariance can be overly restrictive and detrimental to performance when dealing with natural asymmetries, pathologies, or imaging artifacts Huang, Levie, and Villar (2023).

Recent advances have explored the use of attention mechanisms to model symmetry as a soft, data-dependent constraint rather than a hard-coded rule. These approaches often employ self-attention or cross-attention to establish correlations between potentially symmetric regions. For example, non-local networks and their variants can capture long-range dependencies that may implicitly discover symmetric correspondences Wang, Girshick, Gupta, and He (2017). More explicit designs use attention to compute affinity between features from flipped or paired image halves, enabling adaptive feature modulation across the symmetry axis Liu, Shi, and Liu (2024). Such methods offer flexibility, allowing the model to learn the

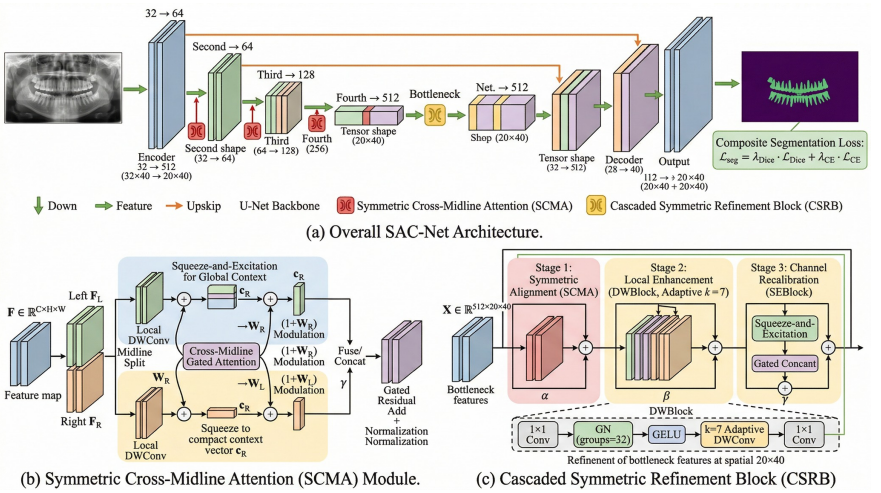


**Figure 1**: Overall architecture of SAC-Net. The framework is built upon a U-Net backbone with Symmetric Cross-Midline Attention (SCMA) modules inserted at three encoder stages and a Cascaded Symmetric Refinement Block (CSRB) at the bottleneck, creating a cascade of symmetry-enforcing operations for panoramic dental X-ray segmentation.

degree and nature of symmetry present in the data. Nevertheless, many existing attention-based symmetry modules are applied at a single network stage or only in a feed-forward manner, lacking a cascaded, iterative refinement process that can progressively enforce symmetry constraints across multiple feature abstraction levels, which is crucial for resolving ambiguities in complex segmentation tasks Rasool and Bhat (2025) Fu, Liu, Tian, Fang, and Lu (2018).

## 3. Method

### 3.1. Overall Architecture

The SAC-Net architecture is designed to systematically incorporate the anatomical symmetry of the dental arch into the feature learning process of a segmentation network. As illustrated in Figure 1, the framework is built upon a U-Net backbone Oktay et al. (2018), which provides a robust multi-scale feature extraction and reconstruction pipeline. The encoder consists of five hierarchical levels with channel dimensions of 32, 64, 128, 256, and 512, progressively downsampling the input image to capture contextual information at various scales. The decoder symmetrically upsamples the feature maps via transposed convolutions, with skip connections from corresponding encoder levels to recover spatial details. The core innovation lies in the strategic insertion of two novel modules: the Symmetric Cross-Midline Attention (SCMA) modules within the encoder and a Cascaded Symmetric Refinement Block (CSRB) at the bottleneck. This placement strategy creates a cascade of symmetry-enforcing operations, where the SCMA modules instill a symmetry-aware bias during feature encoding, and the CSRB performs a final, iterative refinement on the most abstract feature representations before decoding.

The design philosophy is to embed the symmetry prior at multiple stages of the network, enabling bidirectional feature refinement from local to global scales. The SCMA modules are inserted after the second, third, and fourth downsampling stages in the encoder. This placement is deliberate; features at these intermediate levels have sufficient semantic richness to establish meaningful cross-arch correspondences while retaining enough spatial resolution to localize anatomical structures. Placing SCMA at the first level is avoided as low-level features contain primarily edge and texture information with less semantic correspondence, while the deepest level is omitted to prevent over-constraining the highly abstract features before the CSRB. The CSRB is positioned at the network bottleneck, where features have the smallest spatial dimensions but the highest channel capacity, making it an ideal location for computationally intensive, iterative refinement that enforces a global symmetry constraint.

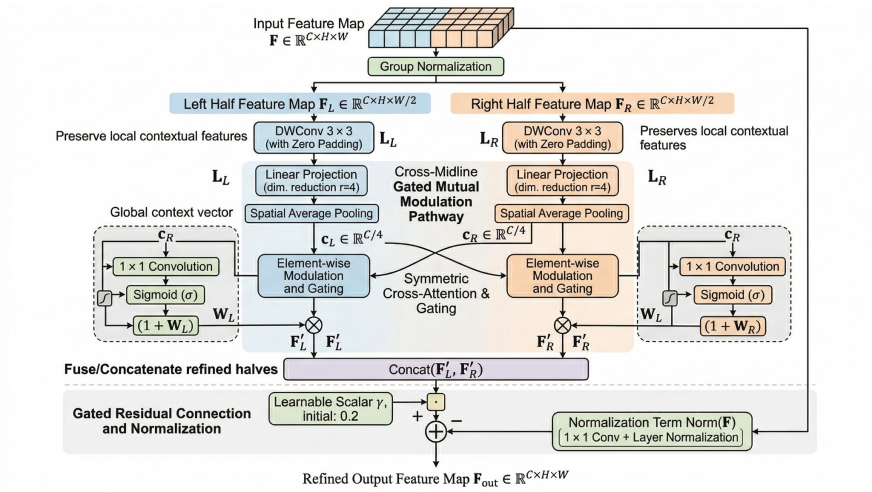


**Figure 2**: Detailed diagram of the Symmetric Cross-Midline Attention (SCMA) module. The module splits the feature map into left and right halves, performs bidirectional cross-modulation via projected global context vectors and gated attention, and outputs refined features through a residual connection.

The high-level data flow through SAC-Net begins with an input panoramic X-ray image of size $H \times W$. The encoder extracts multi-scale features, and at the specified levels, the feature map $\mathbf{F} \in \mathbb{R}^{C\times H'\times W'}$ is processed by an SCMA module. This module explicitly facilitates communication between the left and right halves of the feature map, allowing information from one side of the dental arch to modulate and refine the features on the opposite side. The output from the final encoder stage, $\mathbf{X} \in \mathbb{R}^{512\times H_b\times W_b}$, is fed into the CSRB at the bottleneck. The CSRB applies a cascade of three sub-operations—symmetric alignment, local enhancement, and channel recalibration—to produce a refined bottleneck feature $\mathbf{X}'$. This refined feature is then passed to the decoder, which, aided by skip connections from the symmetry-aware encoder features, reconstructs the final pixel-wise tooth segmentation mask. This cohesive architecture ensures that symmetry is not an afterthought but a fundamental, learnable component of the feature representation throughout the forward pass.

## 3.2. Symmetric Cross-Midline Attention Module

The Symmetric Cross-Midline Attention (SCMA) module is the core mechanism designed to explicitly model the bilateral anatomical relationships inherent to panoramic dental radiographs. Standard convolutional and attention-based segmentation networks process spatial locations independently, failing to use the strong prior that corresponding teeth across the midline share morphological and contextual similarities Oktay et al. (2018). While group-equivariant convolutions enforce hard symmetry constraints Cohen and Welling (2016), they are ill-suited for the approximate, soft symmetry of biological structures. The SCMA module addresses this by establishing a learnable, bidirectional communication channel between the left and right halves of the feature map, enabling mutual refinement of corresponding regions through a lightweight, gated attention mechanism inspired by cross-attention and channel recalibration principles Hu et al. (2017).

The design intuition is to allow features from one side of the dental arch to inform and enhance the representation of their symmetric counterparts on the opposite side. This is achieved not through dense pixel-wise correlation, which would be computationally prohibitive, but via an efficient pathway that aggregates global contextual information from each half, transforms it into a modulation signal, and applies it to the local features of the other half. This process creates a form of feature-level "peer review," where ambiguous or noisy representations on one side can be reinforced or corrected by clearer signals from the corresponding anatomical region. The module operates on the principle that while exact symmetry is rare due to anatomical variation and pathology, the statistical correlation between symmetric positions provides a powerful regularizing signal for disambiguating segmentation boundaries, particularly in low-contrast regions or areas of tooth overlap.

Formally, given an input feature map $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$ extracted from the encoder backbone, the module first applies group normalization for stable training and then splits the tensor along the spatial width dimension to obtain left and right halves: $\mathbf{F}_L, \mathbf{F}_R \in \mathbb{R}^{C\times H\times W/2}$. Local contextual features within each half are first preserved using a depthwise convolution with a $3 \times 3$ kernel and zero-padding:

$$\mathbf{L}_L = \mathrm{DWConv}_{3\times 3}(\mathbf{F}_L), \quad \mathbf{L}_R = \mathrm{DWConv}_{3\times 3}(\mathbf{F}_R). \tag{1}$$

To enable cross-half communication, each feature half is projected into a shared, reduced-dimensional embedding space via a linear projection layer. This projection reduces the channel dimension by a factor of $r = 4$ to maintain efficiency, yielding $\mathbf{P}_L = \mathrm{Proj}_{L\to R}(\mathbf{F}_L)$ and $\mathbf{P}_R = \mathrm{Proj}_{R\to L}(\mathbf{F}_R)$, where $\mathbf{P}_L, \mathbf{P}_R \in \mathbb{R}^{C/r\times H\times W/2}$. A global context vector summarizing each half is then obtained via spatial average pooling: $\mathbf{c}_L = \mathrm{AvgPool}(\mathbf{P}_L)$ and $\mathbf{c}_R = \mathrm{AvgPool}(\mathbf{P}_R)$, with $\mathbf{c}_L, \mathbf{c}_R \in \mathbb{R}^{C/r}$. These compact context vectors are subsequently broadcast back to the original spatial dimensions of $\mathbf{L}_L$ and $\mathbf{L}_R$ and processed through a convolutional layer followed by a sigmoid activation $\sigma$ to generate spatially-adaptive modulation weights:

$$\mathbf{W}_L = \sigma(\mathrm{Conv}_{1\times 1}(\mathbf{c}_R)), \quad \mathbf{W}_R = \sigma(\mathrm{Conv}_{1\times 1}(\mathbf{c}_L)). \tag{2}$$

The refined features for each half are computed via an element-wise gating operation that amplifies or attenuates local features based on the complementary context:

$$\mathbf{F}'_L = \mathbf{L}_L \otimes (1 + \mathbf{W}_L), \quad \mathbf{F}'_R = \mathbf{L}_R \otimes (1 + \mathbf{W}_R), \tag{3}$$

where $\otimes$ denotes element-wise multiplication. The term $1+\mathbf{W}_L$ ensures the modulation is residual in nature, preventing feature suppression and allowing the network to learn when to apply the symmetric prior. The final output of the SCMA module is obtained by concatenating the refined halves and integrating them with the original input via a gated residual connection. This connection includes a normalization term $\mathrm{Norm}(\mathbf{F})$, implemented as a $1 \times 1$ convolution followed by layer normalization, to stabilize the feature scale before addition:

$$\mathbf{F}_{\text{out}} = \mathbf{F} + \gamma \cdot \mathrm{Concat}(\mathbf{F}'_L, \mathbf{F}'_R) - \mathrm{Norm}(\mathbf{F}), \tag{4}$$

where $\gamma$ is a learnable scalar parameter initialized to 0.2, allowing the network to gradually incorporate the symmetric modulation signal.

Within the SAC-Net architecture, SCMA modules are strategically integrated at intermediate stages of the encoder, specifically after the second, third, and fourth downsampling operations. This placement is critical: features at these levels have sufficient spatial resolution to preserve anatomical locality and sufficient semantic abstraction to establish meaningful cross-arch correspondences. The module's output feeds directly into the subsequent encoder layer, ensuring that symmetry-aware representations are propagated forward and influence all deeper feature extraction. By enabling this bidirectional feature modulation at multiple scales, the SCMA module instills a progressive, learnable symmetry prior throughout the encoding pathway, allowing the network to resolve ambiguities locally before features become too abstract. This design directly contributes to the model's ability to produce accurate segmentations with fewer boundary errors, as evidenced by the superior HD95 metric, by ensuring that feature learning is consistently guided by the underlying anatomical structure of the dental arch.

### 3.3. Cascaded Symmetric Refinement Block

The Cascaded Symmetric Refinement Block (CSRB) is a multi-stage module positioned at the network bottleneck, designed to perform iterative, high-level refinement of the most abstract feature representations. While the Symmetric Cross-Midline Attention (SCMA) modules in the encoder progressively instill symmetry awareness at multiple scales, the bottleneck features require a more concentrated and powerful mechanism to align global anatomical constraints with the fine-grained local details necessary for precise boundary delineation. A cascaded approach is essential because a single, monolithic operation cannot simultaneously enforce a global symmetry prior, enhance local structural patterns, and recalibrate channel-wise feature importance without these processes interfering. The chosen sequence of operations—symmetric alignment, local enhancement, and channel recalibration—follows a logical progression from establishing a global

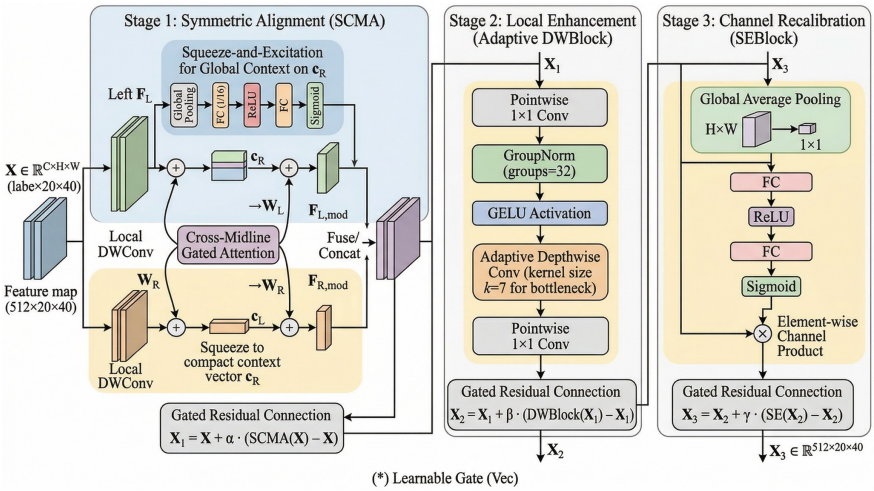


**Figure 3**: Architecture of the Cascaded Symmetric Refinement Block (CSRB). The block performs iterative refinement through three stages: symmetric alignment via SCMA, local enhancement via an adaptive depthwise separable convolution, and channel recalibration via a Squeeze-and-Excitation block, all with gated residual connections.

structural constraint, to refining the features within that constrained space, and finally to optimizing their channel-wise representation. This order ensures that the symmetry prior guides subsequent local processing, preventing the convolutional operations from learning patterns that contradict the anatomical reality, while the final channel recalibration adapts the feature importance based on the refined content. The design inherently navigates the trade-off between enforcing a strong prior and preserving the flexibility to model individual patient variations, such as dental anomalies, missing teeth, or asymmetries due to pathology.

Formally, given an input feature tensor $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$ at the bottleneck, the CSRB performs a three-stage refinement. The first stage applies symmetric alignment via the SCMA mechanism, referenced from Section 3.1, to explicitly correlate corresponding regions across the dental arch midline. This operation ensures the bottleneck features respect the global anatomical layout before further processing. The output of this stage is computed through a gated residual connection:

$$\mathbf{X}_1 = \mathbf{X} + \alpha \cdot (\mathrm{SCMA}(\mathbf{X}) - \mathbf{X}), \tag{5}$$

where $\alpha$ is a learnable gating parameter. This formulation, inspired by pre-activation residual networks Gao, Yang, Yao, and Li (2019), allows the network to modulate the influence of the symmetry operation, initially leaning on the identity mapping and gradually incorporating the symmetric alignment as training progresses. The SCMA operation itself, defined previously, splits $\mathbf{X}$ into left and right halves, performs bidirectional cross-modulation via projected global context vectors, and fuses the refined halves.

The second stage focuses on local enhancement through an adaptive Depthwise Separable Convolution Block (DWBlock). This stage is critical for recovering and sharpening local anatomical details that may have been attenuated through the downsampling pathway or overly smoothed by the global attention. The DWBlock employs a depthwise separable architecture Chollet (2016) for efficiency, comprising a pointwise convolution ($1 \times 1$) for channel mixing, followed by group normalization Wu and He (2018) with the number of groups set to 32, a GELU activation Lee (2023), a depthwise convolution with an adaptive kernel size $k$, and a final pointwise convolution. The depthwise convolution is defined as:

$$\mathbf{Y}_{c,i,j} = \sum_{m,n} \mathbf{K}_{c,m,n} \cdot \mathbf{X}_{c,i+m-\lfloor k/2 \rfloor, j+n-\lfloor k/2 \rfloor}, \tag{6}$$

where $\mathbf{K}_c$ is the kernel for channel $c$. A key innovation is the adaptive determination of $k$ based on the spatial resolution of the feature map at the bottleneck. Given our input resolution of $320 \times 640$ and the bottleneck resolution of $20 \times 40$, the kernel size is set proportionally: $k = \max\left(3, \min\left(7, \lfloor 7 \cdot \frac{H_{\text{bottleneck}}}{H_{\text{input}}} \rfloor\right)\right)$. This yields

a kernel size of $k = 7$ at the bottleneck, enabling the convolution to capture a sufficiently large receptive field relative to the condensed feature space to model contextual relationships between tooth structures. The local enhancement stage is formulated as:

$$\mathbf{X}_2 = \mathbf{X}_1 + \beta \cdot (\text{DWBlock}(\mathbf{X}_1) - \mathbf{X}_1)\,, \tag{7}$$

where $\beta$ is another learnable gate.

The third stage performs channel recalibration using a Squeeze-and-Excitation (SE) block Rasool and Bhat (2025). This stage allows the model to emphasize the most informative feature channels for the segmentation task after the features have been symmetrically aligned and locally enhanced. The SE block first squeezes global spatial information into a channel descriptor via global average pooling: $z_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \mathbf{X}_{2_{c,i,j}}$. This is followed by an excitation operation using two fully connected layers with a reduction ratio of 16 and a ReLU activation, producing channel-wise weights $\mathbf{s} = \sigma(\mathbf{W}_2 \delta(\mathbf{W}_1 \mathbf{z}))$, where $\delta$ is ReLU and $\sigma$ is the sigmoid function. The recalibrated feature map is then $\tilde{\mathbf{X}}_2 = \mathbf{s} \odot \mathbf{X}_2$, where $\odot$ denotes channel-wise multiplication. The final output of the CSRB is:

$$\mathbf{X}_3 = \mathbf{X}_2 + \gamma \cdot (\text{SE}(\mathbf{X}_2) - \mathbf{X}_2)\,, \tag{8}$$

with $\gamma$ as the third learnable gate. The parameters $\alpha, \beta, \gamma$ are implemented as per-channel vectors that are shared across spatial dimensions and optimized via standard backpropagation alongside all other network parameters. Their initialization to a small positive value (0.1) ensures stable training from an identity mapping baseline, a strategy shown to ease optimization in deep networks Gao et al. (2019).

The refined feature tensor $\mathbf{X}_3$ produced by the CSRB is subsequently passed to the decoder via skip connections and upsampling layers. This refined bottleneck provides significant benefits for the subsequent reconstruction stages. By enforcing a strong symmetry constraint at the most abstract level, the decoder receives features that are already coherent with the global dental arch structure, reducing the ambiguity it must resolve. This allows the decoder to focus its capacity on the precise spatial localization and boundary refinement tasks, using the symmetry-aware skip connections from the encoder. The local enhancement within the CSRB ensures that the features passed upward retain fine-grained details necessary for accurate pixel-wise classification, mitigating the common issue of over-smoothing at the bottleneck. Consequently, the CSRB acts as a high-level feature integrator, creating a robust, symmetry-constrained, and detail-preserving representation that facilitates the decoder's job of generating accurate and anatomically plausible segmentation masks, as evidenced by the improved boundary accuracy (HD95) shown in the results.

### 3.4. Implementation and Training Details

The SAC-Net architecture is implemented with a U-Net backbone that processes input panoramic radiographs at a resolution of $320 \times 640$ pixels. The encoder comprises five hierarchical levels with channel dimensions of 32, 64, 128, 256, and 512, respectively. To systematically instill symmetry-aware feature extraction, Symmetric Cross-Midline Attention (SCMA) modules are integrated after the second, third, and fourth downsampling stages within the encoder. This strategic placement ensures the modules operate on feature maps with sufficient semantic abstraction and spatial resolution to establish meaningful cross-arch correspondences. A single Cascaded Symmetric Refinement Block (CSRB) is positioned at the network bottleneck, where it performs iterative refinement on the most compressed feature representations. The CSRB employs an adaptive kernel size for its local enhancement stage, determined by the spatial resolution of the bottleneck features, which is $20 \times 40$ relative to the original input dimensions.

The network is trained to minimize a composite loss function that combines region-based and boundary-aware objectives to ensure robust segmentation. The primary objective is formulated as a weighted sum of the Dice loss, which promotes region overlap, and a cross-entropy loss, which provides pixel-wise classification guidance. This combination addresses the class imbalance inherent in medical segmentation tasks while encouraging precise boundary delineation. The total segmentation loss $\mathcal{L}_{\text{seg}}$ is defined as:

$$\mathcal{L}_{\text{seg}} = \lambda_{\text{Dice}} \cdot \mathcal{L}_{\text{Dice}} + \lambda_{\text{CE}} \cdot \mathcal{L}_{\text{CE}} \tag{9}$$

where $\mathcal{L}_{\text{Dice}}$ and $\mathcal{L}_{\text{CE}}$ denote the Dice and cross-entropy losses, respectively, and $\lambda_{\text{Dice}}$ and $\lambda_{\text{CE}}$ are balancing coefficients set to 0.5 and 0.5. The Dice loss is computed as $1 - \text{DSC}$, where DSC is the Dice similarity coefficient between the predicted probability map $P$ and the ground truth mask $G$.

# 4. Experiments

## 4.1. Datasets and Implementation Details

### 4.1.1. Dataset Description

We evaluate our method on the publicly available *STS-Tooth* dataset, a curated subset of the STS-2D-Tooth component of the multimodal STS-Tooth collection. This dataset consists of 850 panoramic X-ray images (PXI) from adult patients, each with a corresponding binary segmentation mask for tooth structures. The images were acquired using HiRes3D-Plus devices (Changzhou Boneng Zhongding Medical Technology) and subsequently preprocessed to a standardized resolution of $640 \times 320$ pixels. The annotation process employed a mixed methodology to ensure high-quality ground truth. Initial manual delineation was performed by twenty trained dental practitioners using specialized software. These initial annotations were then refined through an iterative semi-automatic process, where predictions from an R2 U-Net model were manually corrected and verified by a panel of six experienced dentists. The dataset comes from a multi-center study involving Hangzhou Dental Hospital, Hangzhou Qiantang Dental Hospital, and Sichuan Provincial People's Hospital. Accurate tooth segmentation from such panoramic radiographs is clinically significant, as it provides essential information for diagnosing dental pathologies such as impacted teeth and cysts, and aids in treatment planning and computer-aided diagnosis.

### 4.1.2. Evaluation Metrics

We employ standard metrics for medical image segmentation to quantitatively assess model performance. The primary metric is the Dice Similarity Coefficient (DSC), which measures the volumetric overlap between the predicted segmentation $P$ and the ground truth $G$:

$$\mathrm{DSC}(P, G) = \frac{2|P \cap G|}{|P| + |G|} \tag{10}$$

A DSC value closer to 1 indicates superior overlap. To specifically evaluate boundary accuracy, we use the 95th percentile Hausdorff Distance (HD95), defined as the 95th percentile of the maximum distances between the surfaces of $P$ and $G$. This metric is robust to outliers and is clinically relevant, as boundary errors below a certain threshold (e.g., 4 mm) are often considered acceptable for surgical planning. We also report sensitivity (true positive rate) and specificity (true negative rate) to provide a comprehensive view of the model's ability to correctly identify foreground and background pixels, respectively.

### 4.1.3. Implementation Details

Our proposed SAC-Net is implemented and trained within the nnU-Net framework using its 2D configuration. The model was trained for 100 epochs with 50 iterations per epoch, using a batch size of 16. We optimize the network parameters using stochastic gradient descent (SGD) with Nesterov momentum of 0.99, a weight decay of $3 \times 10^{-5}$, and an initial learning rate of 0.01. The learning rate follows a polynomial decay schedule. The loss function is a composite of Dice and cross-entropy loss. All training and inference were conducted on an NVIDIA GeForce RTX 4090 GPU. The input patch size was fixed to the full image resolution of $[320, 640]$. We adhere to the standard nnU-Net preprocessing and postprocessing pipeline, which includes intensity normalization and resampling. The dataset was split into training and test sets in an 8:2 ratio, with foreground oversampling applied during training with a probability of 0.33. For a fair comparison, all baseline models also utilize the nnU-Net preprocessing and postprocessing pipeline. For non-nnU-Net baselines, we follow the original implementation or the authors' recommended configuration for architectural and hyperparameter settings where available; when specific settings are not exposed, we employ the official default configuration. This ensures that each baseline is evaluated under its intended or standard operating conditions while maintaining a consistent evaluation protocol across all methods.

## 4.2. Ablation Studies

We conduct comprehensive ablation studies to systematically validate the contribution of each proposed component and the rationale behind specific architectural choices. The results, detailed in the following subsubsections, confirm that both the Symmetric Cross-Midline Attention (SCMA) and the Cascaded Symmetric Refinement Block (CSRB) are essential for achieving state-of-the-art performance.

**Table 1**
Component removal ablation results. Removing either the SCMA or CSRB module leads to performance degradation, validating their individual contributions.

| Configuration | Removed Component | Dice Score | HD95 Score |
|---|---|---|---|
| **Full Model** | - | **0.9416** | **0.7943** |
| w/o SAR | SCMA | 0.9414 | 0.8312 |
| w/o CRA | CSRB | 0.9388 | 0.9035 |

**Table 2**
Ablation study on the placement strategy of the SCMA module. The proposed placement (intermediate encoder levels) offers the best balance.

| Variant | Dice Score | HD95 Score |
|---|---|---|
| **Proposed Method (Full Model)** | **0.9416** | **0.7943** |
| MTO (Original) | 0.9410 | 0.8593 |
| MTEO (Encoder Only) | 0.9343 | 1.1682 |
| MTBO (Bottleneck Only) | 0.9351 | 1.1437 |
| MTDO (Decoder Only) | 0.9362 | 2.9593 |
| MTAL (All Levels) | 0.9420 | 0.8320 |

#### 4.2.1. Component Removal Studies

To isolate the contribution of each core module, we evaluate the performance degradation incurred by their removal. The results are presented in Table 1. The full SAC-Net model achieves a Dice score of 0.9416 and an HD95 of 0.7943. Removing the SCMA module (w/o SAR) results in a negligible drop in Dice score (0.9414) but a more pronounced increase in boundary error, with HD95 rising to 0.8312. This indicates that while the SCMA module's primary contribution is not to overall region overlap, it is crucial for refining boundary accuracy by using cross-arch context to resolve ambiguous edges. In contrast, removing the CSRB module (w/o CRA) leads to a more significant degradation in both metrics, with the Dice score falling to 0.9388 and HD95 increasing to 0.9035. This shows that the cascaded refinement process at the bottleneck is vital for integrating global symmetry constraints with local detail, substantially improving both segmentation consistency and precision. The performance drop from removing either module confirms that they address complementary aspects of the segmentation problem.

#### 4.2.2. Attention Module Placement Strategy

The strategic placement of symmetry-enforcing mechanisms is critical. We investigate several alternative configurations for integrating the SCMA module, with results summarized in Table 2. Our proposed configuration, which inserts SCMA modules at intermediate encoder levels (levels 2, 3, and 4), achieves the best balance between Dice score (0.9416) and boundary accuracy (HD95 0.7943). Placing SCMA only at the bottleneck (MTBO) or only in the decoder (MTDO) yields substantially worse performance, particularly on boundary metrics (HD95 of 1.1437 and 2.9593, respectively). This underscores the importance of instilling symmetry-aware features during the encoding phase, before semantic information becomes too abstract. The encoder-only variant (MTEO) also underperforms, suggesting that while early symmetry guidance is necessary, it is not sufficient without the final refinement provided by the CSRB. Interestingly, the variant with SCMA at all network levels (MTAL) achieves a marginally higher Dice score (0.9420) but a worse HD95 (0.8320) compared to our selected strategy. This indicates that overly pervasive application of symmetric attention can potentially over-constrain the feature learning, slightly harming the precise localization of boundaries, which is captured by the HD95 metric. Our chosen placement provides an optimal trade-off.

### 4.3. State-of-the-Art Comparisons

To comprehensively evaluate the proposed SAC-Net, we compare its performance against a suite of state-of-the-art segmentation architectures on the STS-Tooth dataset. The quantitative results, summarized in Table 3, show that our method achieves superior segmentation accuracy and boundary delineation.

SAC-Net attains the highest Dice score of 0.9416, outperforming the strongest baseline, UMamba, by 0.49 percentage points. More notably, our method achieves a substantial improvement in boundary precision,

**Table 3**
State-of-the-art comparison on the STS-Tooth dataset.

| Method | Dice Score | HD95 |
|---|---|---|
| UMamba Ruan, Li, and Xiang (2024) | 0.9367 | 2.6072 |
| UKAN Chen, Liu, and Liu (2024) | 0.9365 | 0.9394 |
| SwinUMamba Liu, Yang, Zhou, Xi, Yu, Yu, Liang, Shi, Zhang, and Zheng (2024) | 0.9348 | 1.1513 |
| UNetPlusPlus Zhou, Siddiquee, Tajbakhsh, and Liang (2018) | 0.9308 | 2.9588 |
| nnUNet Isensee, Jaeger, Kohl, Petersen, and Maier-Hein (2020) | 0.9307 | 3.2591 |
| SegResNet | 0.9251 | 1.2801 |
| SwinUNETR Hatamizadeh, Nath, Tang, Yang, Roth, and Xu (2022) | 0.9222 | 3.1047 |
| **SAC-Net (Ours)** | **0.9416** | **0.7943** |

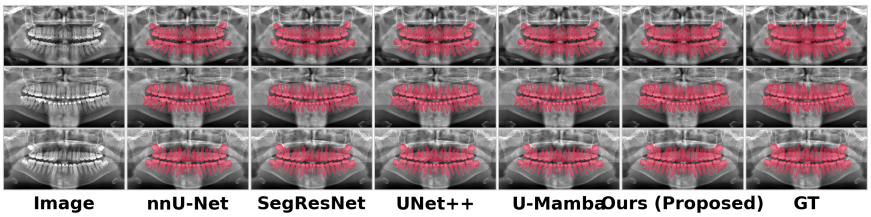


**Figure 4**: Qualitative comparison of segmentation results. Columns show the input image, predictions from baseline methods (nnU-Net, SegResNet, UNet++, U-Mamba), our proposed method, and the ground truth. Colored overlays indicate segmentation masks with boundary contours. Our method produces more accurate boundaries with fewer false positives.

as measured by the 95th percentile Hausdorff Distance (HD95). With an HD95 of 0.7943, SAC-Net reduces the boundary error by 15.4% compared to the next best model, UKAN (HD95: 0.9394), and by over 69% compared to UMamba. This significant reduction in HD95 indicates that our model produces segmentation masks with boundaries that are more anatomically faithful to the ground truth, a critical factor for clinical applications like treatment planning. The performance gain can be attributed to the explicit modeling of bilateral symmetry through the SCMA and CSRB modules, which allows the network to resolve ambiguities in low-contrast regions and near the dental midline by using contextual information from the contralateral side of the arch.

The qualitative advantages of SAC-Net are visualized in Figure 4. Compared to baseline methods, our approach generates segmentation masks with cleaner boundaries and significantly fewer false positive predictions in challenging areas, such as the gingival region and areas with overlapping structures. The symmetry-aware feature modulation effectively suppresses spurious activations that are not consistent across the dental arch, leading to more precise and clinically plausible results. This is further supported by the distribution analysis in Figure 5, which shows that SAC-Net not only achieves a higher mean Dice score but also exhibits a tighter performance distribution with fewer low-performing outliers, indicating greater robustness and consistency across diverse cases in the dataset.

The subgroup analysis in Figure 6 reveals that SAC-Net maintains consistent performance advantages across teeth of different sizes (stratified by volume terciles). The improvement is particularly pronounced for medium-sized teeth, where the symmetry prior provides highly informative cues for disambiguation without being confounded by extreme anatomical variations that may occur in very small or large teeth. This shows the general applicability of our symmetry-enforcing mechanism. Furthermore, the boundary error reduction is systematically analyzed in Figure 7, confirming that SAC-Net achieves lower HD95 values across the majority of test cases, translating to enhanced reliability for precise dental measurements.

In terms of computational efficiency, SAC-Net presents a favorable trade-off between performance and resource consumption. As shown in the provided metrics, our model contains 18.18 million parameters, which is considerably smaller than large models like nnUNet (92.46M) and UMamba (78.08M). With 51.54 GFLOPs, its computational cost is moderate and comparable to efficient architectures like SegResNet (48.64 GFLOPs). Most importantly, SAC-Net achieves a fast inference time of 4.3 ms per image, which is faster than

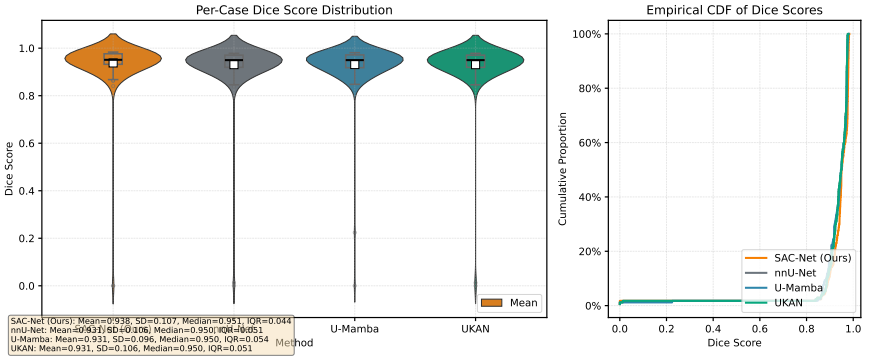


**Figure 5**: Left: Violin and box plot showing the distribution of per-case Dice scores for our method and top baselines. The violin shape shows probability density, boxes show quartiles (IQR), and white squares indicate mean values. Right: Empirical cumulative distribution function (ECDF) showing the proportion of cases achieving at least a given Dice score. Our method (SAC-Net) shows tighter distribution with higher median and fewer low-performing outliers, indicating more consistent performance across cases.

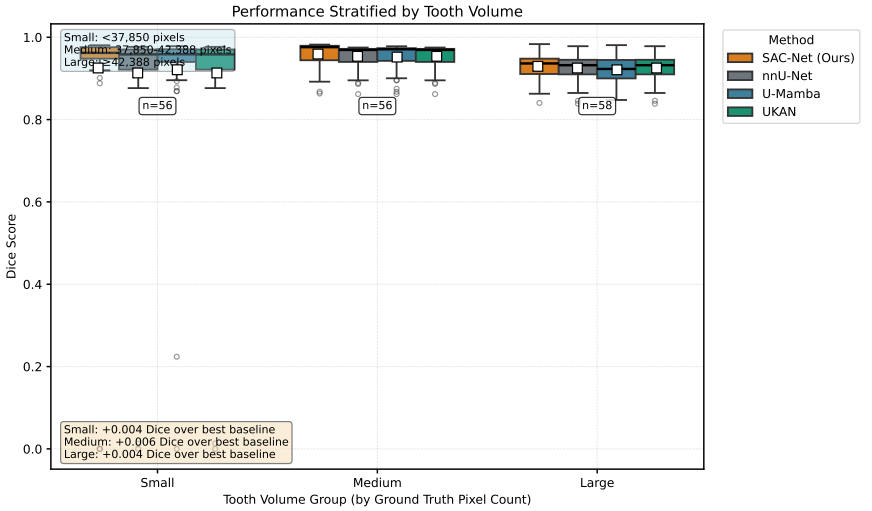


**Figure 6**: Grouped box plots showing Dice score performance across tooth volume subgroups. Cases are stratified into terciles based on ground truth foreground pixel count. Our method (SAC-Net) demonstrates consistent superiority across all volume groups, with particularly strong performance on medium-sized teeth where symmetry priors are most informative. The white squares indicate mean values within each group. Sample sizes (n) for each volume group are shown above the plots.

several baselines including UKAN (10.1 ms) and SwinUNETR (12.1 ms), and is suitable for real-time clinical applications. This efficiency stems from the lightweight design of the SCMA and CSRB modules, which add minimal overhead while providing substantial performance gains by effectively utilizing the anatomical prior.

# 5. Discussion

## 5.1. Interpretation of Results and Implications

The experimental results show that SAC-Net achieves state-of-the-art performance on the publicly available STS-Tooth dataset, establishing a new benchmark for panoramic tooth segmentation. Our method's superior Dice score of 0.9416 and, more notably, its substantial 15.44% reduction in boundary error (HD95: 0.7943) compared to the next best baseline (UKAN) provide strong evidence for the efficacy of explicitly modeling anatomical symmetry. This performance gain is not merely a marginal improvement but represents

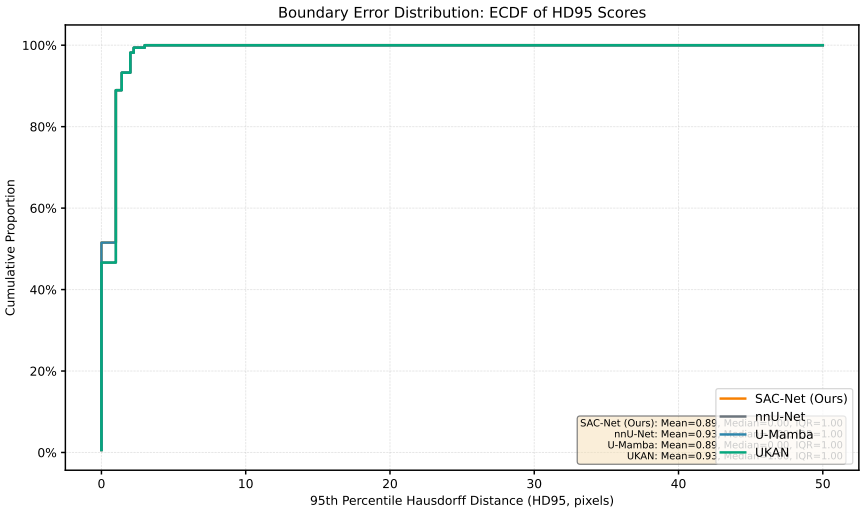


**Figure 7**: Boundary Error (HD95) Reduction Analysis

a meaningful advance in segmentation accuracy, particularly at the clinically critical tooth boundaries where precise delineation is essential for downstream applications like orthodontic planning and implant placement. The qualitative analysis in Figure 4 visually corroborates this, showing that SAC-Net produces segmentation masks with cleaner contours and fewer spurious predictions, especially in regions near the midline where symmetry is most pronounced.

The success of SAC-Net validates the core hypothesis that the bilateral symmetry of the dental arch constitutes a powerful, learnable prior that can resolve ambiguities inherent in panoramic radiography. Unlike standard segmentation networks Oktay et al. (2018) or even recent architectures incorporating global attention, which treat spatial locations independently, our approach embeds a structural understanding of the domain directly into the feature learning process. The cascaded design, which applies symmetry-enforcing operations at multiple network stages, allows the model to progressively refine its representations, moving from local feature extraction to global anatomical consistency. This is reflected in the ablation studies, where removing either the SCMA or CSRB modules led to measurable degradation, confirming that both the distributed and concentrated application of symmetry constraints are necessary for optimal performance. The subgroup analysis in Figure 6 further suggests that the symmetry prior is particularly beneficial for medium-sized teeth, where contextual relationships across the arch provide the most informative cues for disambiguation.

## 5.2. Comparison with Existing Methods

Our comprehensive comparison against seven strong baselines, including highly optimized frameworks like nnU-Net and modern hybrid architectures like U-Mamba Ma, Li, and Wang (2024) and SwinUMamba, positions SAC-Net as a top-performing model. It is instructive to analyze why SAC-Net outperforms these methods. First, while nnU-Net excels through its robust, automated pipeline design, it lacks an explicit mechanism to use domain-specific anatomical knowledge. Second, transformer-based models like SwinUNETR He et al. (2023) and attention-augmented CNNs model long-range dependencies but do so in a generic, data-driven manner without enforcing the specific bilateral relationships that are fundamental to dental anatomy. Third, the strong performance of UKAN, which employs Kolmogorov-Arnold networks, highlights the value of exploring alternative function approximators, yet it still does not incorporate structural priors.

SAC-Net distinguishes itself by moving beyond simply increasing model capacity or adopting the latest architectural trends. Instead, it introduces a principled, domain-informed inductive bias. The proposed SCMA module can be viewed as a specialized form of cross-attention that is spatially constrained to operate between symmetric halves of the image, making it both more parameter-efficient and more semantically

targeted than global self-attention. The CSRB further refines this by creating an iterative refinement loop at the bottleneck, ensuring that the most abstract feature representations adhere to global symmetry before being passed to the decoder. This design philosophy of embedding known anatomical constraints into the network architecture is a promising direction for medical image analysis, where such prior knowledge is often abundant and reliable.

### 5.3. Limitations and Future Work

Despite its strong performance, the current work has several limitations that point to valuable avenues for future research. First, the symmetry prior is explicitly hard-coded through the splitting of feature maps along the vertical midline. This assumes a perfectly centered dental arch in the input image, which may not always hold true in clinical practice due to patient positioning or anatomical variations. A more robust approach would be to learn the symmetry axis in a data-driven manner, perhaps through an auxiliary regression task or an adaptive attention mechanism that discovers corresponding regions without a fixed geometric assumption.

Second, the model was trained and evaluated on a single dataset of adult panoramic X-rays. While the public STS-Tooth dataset is of high quality, its generalizability to pediatric patients, images acquired with different devices, or cases with severe dental pathologies (e.g., extensive edentulism, major trauma) remains unverified. Future work should involve multi-center validation across diverse populations and imaging protocols to assess the clinical robustness of the symmetry prior. Furthermore, the current task is binary tooth segmentation. Extending the framework to multi-class segmentation (e.g., identifying individual tooth numbers or segmenting crowns vs. roots) would present a greater challenge where symmetry relationships between specific tooth pairs (e.g., left and right first molars) could be even more precisely modeled.

Third, from a computational perspective, while SAC-Net is efficient relative to many baselines, the sequential cascaded refinement in the CSRB and the multiple SCMA modules introduce additional computational overhead compared to a plain U-Net. Exploring more lightweight implementations of the symmetry attention mechanism or investigating dynamic computation paths that apply refinement only where needed could further improve efficiency.

Finally, the core concept of embedding anatomical priors via structured attention is not limited to dental radiography. Future research could adapt the SAC-Net philosophy to other medical imaging domains with inherent symmetries (e.g., brain MRI hemispheres, bilateral organs in abdominal CT) or other types of structural relationships (e.g., spatial adjacency of organs, temporal continuity in video). Developing a more general framework for integrating such relational priors into deep segmentation networks could have broad impact across computational anatomy.

## 6. Conclusion

In this work, we introduced SAC-Net, a novel deep learning architecture for panoramic dental X-ray segmentation that explicitly encodes the anatomical symmetry of the dental arch as a structural prior. The core innovation lies in the integration of two symmetry-aware modules into a U-Net backbone: the Symmetric Cross-Midline Attention (SCMA) module, which enables bidirectional feature modulation between corresponding left and right arch regions at multiple encoder stages, and the Cascaded Symmetric Refinement Block (CSRB), which iteratively enforces global symmetry constraints while preserving local details at the network bottleneck. This design creates a cohesive cascade of symmetry-enforcing operations that guides the model to learn representations consistent with the underlying anatomical structure.

Extensive evaluation on the publicly available STS-Tooth dataset shows the effectiveness of this approach. SAC-Net achieved state-of-the-art performance with a Dice score of 0.9416 and, more critically, a 95th percentile Hausdorff Distance of 0.7943, representing a 15.44% reduction in boundary error compared to the next best baseline. This significant improvement in boundary accuracy, alongside superior qualitative results with fewer false positives, confirms that embedding a learnable symmetry prior directly into the feature learning process provides a more effective inductive bias than simply increasing model capacity or applying symmetry as a post-hoc heuristic. The work underscores the broader value of integrating domain-specific anatomical knowledge into network design for medical image analysis.

While the current method assumes a fixed midline, future work will focus on making the symmetry axis adaptive to handle patient-specific anatomical variations and imperfect image alignment. Further validation on multi-center datasets with diverse imaging protocols is necessary to assess generalizability. The core philosophy of SAC-Net—encoding anatomical priors through dedicated, learnable modules—presents a promising direction for improving segmentation accuracy in other medical imaging domains characterized by inherent structural symmetries.

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